Business Problem Statement

Uber is experiencing rapid growth in ride bookings but faces challenges such as high cancellation rates, inconsistent driver performance, and limited understanding of customer booking behavior by analyzing the 70,000+ booking records to answer :

How can ride-booking operations be analyzed to improve booking completion rates, revenue optimization, and customer and driver experience?

Deliverables

1. Data Preparation & Modeling (Python) Clean and transform the raw Uber booking dataset. Handle missing values in columns like Driver_Ratings, Customer_Rating, and Incomplete_Rides_Reason.
2. Data Analysis (SQL) Organize the booking data into a structured format and run queries to extract operational insights including vehicle type performance scorecards, cancellation rate breakdowns by source and location, incomplete ride root cause analysis, revenue by pickup zone and payment method, and the relationship between turnaround time (V_TAT, C_TAT) and booking outcomes.
3. Visualization & Insights (Tableau) Build an interactive Power BI dashboard analyzing the Uber booking dataset, highlighting booking completion trends, revenue by vehicle type and location, driver and customer rating patterns, cancellation and incomplete ride breakdowns, and the impact of time-of-day and ride distance on booking success and revenue
4. Report and Presentation Write a clear project report summarizing key findings such as peak demand time slots, top cancellation reasons, high-revenue pickup zones, and driver performance gaps. Prepare a presentation with actionable recommendations for reducing cancellations, improving driver allocation, optimizing pricing by vehicle type, and enhancing customer experience.
5. GitHub Repository Include all Python scripts for data cleaning and feature engineering, SQL queries for business insights, and Power BI dashboard files in a well-structured repository with a detailed README explaining the problem, approach, and key findings.